Stochastic Processes — 1060429

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Throughout, a Brownian motion (BM) is standard ($B_0 = 0$) with continuous paths, and $\mathcal{F}_t$ denotes the filtration in use. Apparent typos are pointed out where they occur.

Exercise 2

Conditional expectation

1.

(a) Let $Z = \mathbb{E}[X|Y]$. By conditional Jensen, $Z^2 \le \mathbb{E}[X^2|Y]$, so $Z \in L^2$. If $\mathbb{E}[h(Y)^2] = \infty$ the right-hand side is infinite (as $X \in L^2$). Otherwise $V = Z - h(Y) \in L^2(\sigma(Y))$, and by Problem 2(b) (the product $V(X - Z)$ is integrable by Cauchy–Schwarz)

$$\mathbb{E}[V(X - Z)] = \mathbb{E}\big[V\,\mathbb{E}[X - Z\,|\,Y]\big] = 0.$$

Hence $\mathbb{E}(X - h(Y))^2 = \mathbb{E}(X - Z)^2 + \mathbb{E}V^2 + 2\mathbb{E}[V(X - Z)] \ge \mathbb{E}(X - Z)^2$.

(b) $L^2(\Omega, \sigma(Y), \mathbb{P})$ is a closed subspace of $L^2(\Omega, \mathcal{F}, \mathbb{P})$. As shown, $Z = \mathbb{E}[X|Y]$ lies in it, and $\mathbb{E}[V(X - Z)] = 0$ for every $V \in L^2(\sigma(Y))$, i.e. $X - Z \perp L^2(\sigma(Y))$. These two properties characterise the orthogonal projection. (Equivalently, by (a): every $\sigma(Y)$-measurable V is of the form $h(Y)$, so Z is the closest point.)

2.

(a) ($\Rightarrow$) Let $Z = \mathbb{E}[X|\mathcal{G}]$. For $W = \mathbf{1}_A$, $A \in \mathcal{G}$, the identity is the definition. By linearity it holds for simple W. For W with $|W| \le c$ a.s., take simple $\mathcal{G}$-measurable $W_n \to W$ with $|W_n| \le c$. Then $|W_n Z| \le c|Z|$ and $|W_n X| \le c|X|$, and dominated convergence gives $\mathbb{E}[WZ] = \mathbb{E}[WX]$.

($\Leftarrow$) Taking $W = \mathbf{1}_A$ gives $\mathbb{E}[Z\mathbf{1}_A] = \mathbb{E}[X\mathbf{1}_A]$ for $A \in \mathcal{G}$. Integrability of Z: with $W_n = \operatorname{sgn}(Z)\mathbf{1}_{\{|Z|\le n\}}$ we get $\mathbb{E}[|Z|\mathbf{1}_{\{|Z|\le n\}}] = \mathbb{E}[W_n X] \le \mathbb{E}|X|$. Letting $n \to \infty$ gives $\mathbb{E}|Z| \le \mathbb{E}|X|$ by monotone convergence. Hence $Z = \mathbb{E}[X|\mathcal{G}]$.

(b) First let $X, Y \ge 0$ and $Z = \mathbb{E}[X|\mathcal{G}] \ge 0$. For $A \in \mathcal{G}$, $W = \mathbf{1}_A \min(Y, n)$ is bounded and $\mathcal{G}$-measurable, so $\mathbb{E}[\mathbf{1}_A \min(Y, n)Z] = \mathbb{E}[\mathbf{1}_A \min(Y, n)X]$. By monotone convergence, $\mathbb{E}[\mathbf{1}_A YZ] = \mathbb{E}[\mathbf{1}_A YX] < \infty$. So YZ is integrable, $\mathcal{G}$-measurable, and equals $\mathbb{E}[XY|\mathcal{G}]$.

In general write $X = X^+ - X^-$ and $Y = Y^+ - Y^-$. Each $X^\pm Y^\pm \le |XY|$ is integrable, and the positive case gives $\mathbb{E}[X^\pm Y^\pm|\mathcal{G}] = Y^\pm \mathbb{E}[X^\pm|\mathcal{G}]$. Summing with signs and using linearity,

$$\mathbb{E}[XY|\mathcal{G}] = (Y^+ - Y^-)(\mathbb{E}[X^+|\mathcal{G}] - \mathbb{E}[X^-|\mathcal{G}]) = Y\,\mathbb{E}[X|\mathcal{G}].$$

3.

$\mathcal{G}$ is generated by the partition $I_1 = [0, \frac{1}{4}]$, $I_2 = (\frac{1}{4}, \frac{2}{3}]$, $I_3 = (\frac{2}{3}, 1]$, so $\mathcal{G} = \{\bigcup_{i\in S} I_i : S \subseteq \{1, 2, 3\}\}$. Let $Z = \sum_i a_i \mathbf{1}_{I_i}$ with $a_i = \mathrm{Leb}(I_i)^{-1} \int_{I_i} \omega^2 d\omega$. Then Z is $\mathcal{G}$-measurable, and $\mathbb{E}[Z\mathbf{1}_{I_i}] = \mathbb{E}[X\mathbf{1}_{I_i}]$ for each i, hence for every $A \in \mathcal{G}$ by additivity. By uniqueness $\mathbb{E}[X|\mathcal{G}] = Z$ a.s. The coefficients are

$$a_1 = 4 \cdot \tfrac{1}{3\cdot 64} = \tfrac{1}{48}, \qquad a_2 = \tfrac{12}{5} \cdot \tfrac{1}{3}\Big(\tfrac{8}{27} - \tfrac{1}{64}\Big) = \tfrac{12}{5} \cdot \tfrac{485}{5184} = \tfrac{97}{432}, \qquad a_3 = 3 \cdot \tfrac{1}{3}\Big(1 - \tfrac{8}{27}\Big) = \tfrac{19}{27}.$$

4.

Remark. Y must be allowed to take the value $+\infty$. For instance, if $\mathcal{G} = \{\emptyset, \Omega\}$ then $\int_\Omega Y\, d\mathbb{P} = \mathbb{E}X = \infty$ forces $Y \equiv \infty$. We prove the statement for $[0, \infty]$-valued Y.

Existence. For $M \in \mathbb{N}$ let $X_M = \min(X, M)$ and $Y_M = \mathbb{E}[X_M|\mathcal{G}]$. Since $0 \le X_M \le X_{M+1}$, monotonicity gives $0 \le Y_M \le Y_{M+1}$ a.s. Outside a null set the sequence is non-decreasing, and $Y := \lim_M Y_M \in [0, \infty]$ is $\mathcal{G}$-measurable. For $A \in \mathcal{G}$, monotone convergence gives

$$\int_A Y = \lim_M \int_A Y_M = \lim_M \int_A X_M = \int_A X.$$

Uniqueness. Let Y, Y' be two such variables and $A_n = \{Y' > Y,\ Y \le n\} \in \mathcal{G}$. Then $\int_{A_n} Y' = \int_{A_n} X = \int_{A_n} Y \le n < \infty$, so $\int_{A_n} (Y' - Y) = 0$ with $Y' - Y > 0$ on A_n. Hence $\mathbb{P}(A_n) = 0$, and taking the union over n gives $\mathbb{P}(Y' > Y, Y < \infty) = 0$. Since $Y' > Y$ forces $Y < \infty$, we get $\mathbb{P}(Y' > Y) = 0$. By symmetry $\mathbb{P}(Y > Y') = 0$.

5. Take $\Omega = \{a, b, c\}$, $\mathcal{F} = 2^\Omega$, $\mathbb{P}$ uniform, $X = \mathbf{1}_{\{a\}}$, $\mathcal{G} = \sigma(\{a\})$, $\mathcal{H} = \sigma(\{c\})$. Then $\mathbb{E}[X|\mathcal{G}] = X$ and

$$\mathbb{E}[\mathbb{E}[X|\mathcal{G}]|\mathcal{H}] = \tfrac{1}{2}\mathbf{1}_{\{a,b\}}, \qquad \mathbb{E}[X|\mathcal{H}] = \tfrac{1}{2}\mathbf{1}_{\{a,b\}}, \qquad \mathbb{E}[\mathbb{E}[X|\mathcal{H}]|\mathcal{G}] = \tfrac{1}{2}\mathbf{1}_{\{a\}} + \tfrac{1}{4}\mathbf{1}_{\{b,c\}}.$$

At c (which has probability $\frac{1}{3}$) the first equals 0 and the second equals $\frac{1}{4}$.

Martingales

1. ($\Rightarrow$) For $A \in \mathcal{F}_s$, $\mathbb{E}[X_t \mathbf{1}_A] = \mathbb{E}[\mathbb{E}[X_t|\mathcal{F}_s]\mathbf{1}_A] \ge \mathbb{E}[X_s \mathbf{1}_A]$. ($\Leftarrow$) Let $Z = \mathbb{E}[X_t|\mathcal{F}_s] - X_s$, which is $\mathcal{F}_s$-measurable with $\mathbb{E}[Z\mathbf{1}_A] \ge 0$ for all $A \in \mathcal{F}_s$. With $A = \{Z < 0\}$ we get $\mathbb{E}[Z\mathbf{1}_{\{Z<0\}}] \ge 0$, so $\mathbb{P}(Z < 0) = 0$, i.e. $\mathbb{E}[X_t|\mathcal{F}_s] \ge X_s$.

2. (The index set in $(X_t, \mathcal{F}_t^X)_{t \in X}$ should be T.) ($\Leftarrow$) Take $\mathcal{F}_t = \mathcal{F}_t^X$. ($\Rightarrow$) Since X is $(\mathcal{F}_t)$-adapted, $\mathcal{F}_t^X \subseteq \mathcal{F}_t$, and X is $\mathcal{F}^X$-adapted and integrable. For $s < t$, by the tower property,

$$\mathbb{E}[X_t|\mathcal{F}_s^X] = \mathbb{E}[\mathbb{E}[X_t|\mathcal{F}_s]|\mathcal{F}_s^X] = \mathbb{E}[X_s|\mathcal{F}_s^X] = X_s.$$

3. (Doob's theorem; (a) should read $\mathbb{E}|X_{\tau_2}| < \infty$.) Here $\mathcal{F}_\infty \supseteq \mathcal{F}_n$ for all n, and the martingale property gives $X_n = \mathbb{E}[X_\infty|\mathcal{F}_n]$. A stopping time is $\tau : \Omega \to \mathbb{N}_\infty$ with $\{\tau = n\} \in \mathcal{F}_n$, $n \in \mathbb{N}$, and $\mathcal{F}_\tau = \{A \in \mathcal{F}_\infty : A \cap \{\tau = n\} \in \mathcal{F}_n\ \forall n \in \mathbb{N}\}$.

Claim: for every stopping time τ, $X_\tau = \mathbb{E}[X_\infty|\mathcal{F}_\tau]$.

$X_\tau = \sum_{n \in \mathbb{N}_\infty} X_n \mathbf{1}_{\{\tau=n\}}$ is $\mathcal{F}_\tau$-measurable, since $\{X_\tau \in B\} \cap \{\tau = n\} = \{X_n \in B\} \cap \{\tau = n\} \in \mathcal{F}_n$. By conditional Jensen, $|X_n| \le \mathbb{E}[|X_\infty||\mathcal{F}_n]$, so

$$\mathbb{E}|X_\tau| \le \sum_n \mathbb{E}[|X_\infty|\mathbf{1}_{\{\tau=n\}}] = \mathbb{E}|X_\infty|.$$

For $A \in \mathcal{F}_\tau$, Fubini (justified by the last bound) gives

$$\mathbb{E}[X_\tau \mathbf{1}_A] = \sum_n \mathbb{E}[X_n \mathbf{1}_{A \cap \{\tau=n\}}] = \sum_n \mathbb{E}[X_\infty \mathbf{1}_{A \cap \{\tau=n\}}] = \mathbb{E}[X_\infty \mathbf{1}_A],$$

because $A \cap \{\tau = n\} \in \mathcal{F}_n$ (trivially for $n = \infty$).

(a) $\mathbb{E}|X_{\tau_2}| \le \mathbb{E}|X_\infty| < \infty$.

(b) If $A \in \mathcal{F}_{\tau_1}$ then $A \cap \{\tau_2 \le n\} = A \cap \{\tau_1 \le n\} \cap \{\tau_2 \le n\} \in \mathcal{F}_n$, so $\mathcal{F}_{\tau_1} \subseteq \mathcal{F}_{\tau_2}$. By the claim and the tower property,

$$\mathbb{E}[X_{\tau_2}|\mathcal{F}_{\tau_1}] = \mathbb{E}[\mathbb{E}[X_\infty|\mathcal{F}_{\tau_2}]|\mathcal{F}_{\tau_1}] = \mathbb{E}[X_\infty|\mathcal{F}_{\tau_1}] = X_{\tau_1}.$$

4. $\mathcal{F}_\tau = \{A : A \cap \{\tau \le u\} \in \mathcal{F}_u \ \forall u\}$. From discrete time we use: if $(Z_g, \mathcal{F}_g)_{g\in G}$ is a supermartingale on a discrete set $G \subset [0,\infty)$ and $\sigma \le \rho \le c$ are G-valued stopping times, then $\mathbb{E}[Z_\rho|\mathcal{F}_\sigma] \le Z_\sigma$, and the stopped process $(Z_{g\wedge\rho})_{g\in G}$ is a supermartingale.

Lemma. Let Y_n be integrable and $\mathcal{H}_n$-measurable with $\mathbb{E}[Y_m\mathbf{1}_B] \le \mathbb{E}[Y_n\mathbf{1}_B]$ for all $m \le n$ and $B \in \mathcal{H}_n$, and suppose $L = \sup_n \mathbb{E}Y_n < \infty$. Then (Y_n) is uniformly integrable.

Proof. Taking $B = \Omega$ shows $\mathbb{E}Y_n \uparrow L$. From $Y_n \ge \mathbb{E}[Y_1|\mathcal{H}_n] \ge -\mathbb{E}[Y_1^-|\mathcal{H}_n]$ we get $\mathbb{E}Y_n^- \le \mathbb{E}Y_1^-$, so $\sup_n \mathbb{E}|Y_n| \le L + 2\mathbb{E}|Y_1| =: K$. Given $\varepsilon > 0$ pick m with $\mathbb{E}Y_m \ge L - \varepsilon$. For $n \ge m$, since $\{Y_n \le \lambda\}, \{Y_n < -\lambda\} \in \mathcal{H}_n$,

$$\begin{aligned}\mathbb{E}[|Y_n|; |Y_n| > \lambda] &= \mathbb{E}Y_n - \mathbb{E}[Y_n; Y_n \le \lambda] - \mathbb{E}[Y_n; Y_n < -\lambda] \\ &\le \mathbb{E}Y_m + \varepsilon - \mathbb{E}[Y_m; Y_n \le \lambda] - \mathbb{E}[Y_m; Y_n < -\lambda] \le \mathbb{E}[|Y_m|; |Y_n| > \lambda] + \varepsilon.\end{aligned}$$

As $\mathbb{P}(|Y_n| > \lambda) \le K/\lambda$ and Y_m is integrable, the right side is $\le 2\varepsilon$ for λ large, uniformly in $n \ge m$. The finitely many $n < m$ are handled trivially. □

(a) Fix $s < t$ and let $G_n = 2^{-n}\mathbb{N} \cup \{s,t\}$; these sets are nested. Let $\tau_n = \inf\{g \in G_n : g > \tau\}$, with $\tau_n = \infty$ if $\tau = \infty$. Then $\{\tau_n \le g\} = \{\tau < g\} \in \mathcal{F}_g$, so τ_n is a G_n-valued stopping time, and it is also a stopping time for the grid $G_{n'}$, $n' \ge n$. Moreover $\tau < \tau_n \le \tau + 2^{-n}$ and $\tau_n \downarrow \tau$.

For $r \in \{s,t\}$, $X_{r\wedge\tau_n} = \sum_{g\in G_n\cap[0,r]} X_g\mathbf{1}_{\{r\wedge\tau_n = g\}}$ is $\mathcal{F}_r$-measurable. By right-continuity $X_{r\wedge\tau_n} \to X_{r\wedge\tau}$, so $X_{r\wedge\tau}$ is $\mathcal{F}_r$-measurable. For $u < r$, $\{X_{r\wedge\tau} \in B\} \cap \{\tau \le u\} = \{X_{u\wedge\tau} \in B\} \cap \{\tau \le u\} \in \mathcal{F}_u$, hence $X_{r\wedge\tau}$ is $\mathcal{F}_{r\wedge\tau}$-measurable.

Since the stopped discrete process is a supermartingale and $s, t \in G_n$, for $A \in \mathcal{F}_s$,

$$\mathbb{E}[X_{t\wedge\tau_n}\mathbf{1}_A] \le \mathbb{E}[X_{s\wedge\tau_n}\mathbf{1}_A]. \tag{$*$}$$

Let $Y_n = X_{r\wedge\tau_n}$ and let $\mathcal{H}_n$ be the σ-algebra of $r \wedge \tau_n$ on the grid G_n. For $m \le n$, $r\wedge\tau_n \le r\wedge\tau_m \le r$ are G_n-stopping times, so $\mathbb{E}[Y_m|\mathcal{H}_n] \le Y_n$. Also $\mathbb{E}Y_n \le \mathbb{E}X_0$. By the Lemma, (Y_n) is UI. Together with a.s. convergence this gives $X_{r\wedge\tau_n} \to X_{r\wedge\tau}$ in L^1, so $X_{r\wedge\tau}$ is integrable. Passing to the limit in $(*)$ gives $\mathbb{E}[X_{t\wedge\tau}\mathbf{1}_A] \le \mathbb{E}[X_{s\wedge\tau}\mathbf{1}_A]$ for $A \in \mathcal{F}_s$. By Problem 1 (with reversed inequality), $(X_{t\wedge\tau}, \mathcal{F}_t)$ is a supermartingale.

Since $\mathcal{F}_{s\wedge\tau} \subseteq \mathcal{F}_s$ (if $A \in \mathcal{F}_{s\wedge\tau}$ then $A = A \cap \{s\wedge\tau \le s\} \in \mathcal{F}_s$), $(\mathcal{F}_{t\wedge\tau})$ is a filtration to which $X_{t\wedge\tau}$ is adapted. The same inequality for $A \in \mathcal{F}_{s\wedge\tau}$ shows that $(X_{t\wedge\tau}, \mathcal{F}_{t\wedge\tau})$ is a supermartingale.

(b) Applying (a) to X and $-X$, $(X_{t\wedge\tau})$ is a martingale, so $\mathbb{E}X_{t\wedge\tau} = \mathbb{E}X_0$. On $\{\tau < \infty\}$, $X_{t\wedge\tau} = X_\tau$ for $t \ge \tau$. On $\{\tau = \infty\}$, X_τ must be given a meaning. We set $X_\tau := \lim_t X_{t\wedge\tau}$, which exists a.s. by Problem 5(a) because $\sup_t \mathbb{E}X_{t\wedge\tau}^+ \le \mathbb{E}Y$. (If $\tau < \infty$ a.s. this is not needed.) Then $|X_\tau| \le Y$, so X_τ is integrable, and dominated convergence gives $\mathbb{E}X_\tau = \lim_t \mathbb{E}X_{t\wedge\tau} = \mathbb{E}X_0$.

5.

(a) Let D be the non-negative dyadic rationals and $F_k = D_k \cap [0,k]$. For $a < b$ let U_k be the number of upcrossings of $[a,b]$ by $(X_t)_{t\in F_k}$. By Doob's upcrossing inequality for the discrete submartingale on the finite set F_k,

$$(b-a)\mathbb{E}U_k \le \mathbb{E}(X_k - a)^+ \le \sup_t \mathbb{E}X_t^+ + |a|.$$

U_k increases to the number U_D of upcrossings by $(X_t)_{t\in D}$, so by monotone convergence $\mathbb{E}U_D < \infty$. Hence a.s. $U_D[a,b] < \infty$ for all rational $a < b$, and $\lim_{D\ni t\to\infty} X_t$ exists in $[-\infty,\infty]$ (otherwise $\liminf < a < b < \limsup$ would give infinitely many upcrossings).

By right-continuity, if $X_t > c$ for arbitrarily large real t then $X_d > c$ for arbitrarily large $d \in D$. So $\limsup_{t\to\infty} X_t = \limsup_{D\ni t\to\infty} X_t$, and similarly for $\liminf$. Thus $X = \lim_{t\to\infty} X_t$ exists a.s. Since $\mathbb{E}|X_t| = 2\mathbb{E}X_t^+ - \mathbb{E}X_t \le 2\sup_t \mathbb{E}X_t^+ - \mathbb{E}X_0$, Fatou's lemma gives $\mathbb{E}|X| < \infty$.

(b)

Remark. Right-continuity must be assumed here as well. Let $U \sim U[0,1]$ and $X_t = \mathbf{1}_{\{t-\lfloor t \rfloor = U\}}$. Then $X_t = 0$ a.s. for each t, so X is an L^2-bounded martingale, but every path equals 1 at $U + k$ for all $k \in \mathbb{N}$ and does not converge.

Assume right-continuity. Since $\sup \mathbb{E}X_t^+ \le \sup(\mathbb{E}X_t^2)^{1/2} < \infty$, (a) gives $X_t \to X$ a.s. For $s < t$, by Conditional expectation, Problem 2(b), $\mathbb{E}[X_s(X_t - X_s)] = \mathbb{E}[X_s \mathbb{E}[X_t - X_s|\mathcal{F}_s]] = 0$. Hence $\mathbb{E}(X_t - X_s)^2 = \mathbb{E}X_t^2 - \mathbb{E}X_s^2$. So $\mathbb{E}X_t^2$ is non-decreasing and bounded, and $\mathbb{E}(X_t - X_s)^2 \to 0$ as $s, t \to \infty$. By completeness of L^2, $X_t \to X'$ in L^2. Along a sequence $t_n \to \infty$ a subsequence converges a.s. to X', so $X' = X$ a.s. and $X \in L^2$. Finally, for $u > t$, $X_t = \mathbb{E}[X_u|\mathcal{F}_t]$ and

$$\|\mathbb{E}[X_u|\mathcal{F}_t] - \mathbb{E}[X|\mathcal{F}_t]\|_1 \le \|X_u - X\|_1 \to 0,$$

so $\mathbb{E}[X|\mathcal{F}_t] = X_t$ a.s.

6.

(a) The argument works for every $\sigma \in \mathbb{R}$, which we need in (b). Let $M_t(\sigma) = \exp(\sigma B_t - \sigma^2 t/2)$. It is adapted, and integrable since $\mathbb{E}e^{\sigma B_t} = e^{\sigma^2 t/2}$. Since $B_t - B_s$ is independent of $\mathcal{F}_s$,

$$\mathbb{E}[M_t|\mathcal{F}_s] = M_s \, \mathbb{E}\big[e^{\sigma(B_t - B_s)}\big] e^{-\sigma^2(t-s)/2} = M_s.$$

(b) Writing $x = B_t$, $e^{\sigma x - \sigma^2 t/2} = e^{\sigma x} e^{-\sigma^2 t/2} = \sum_k \frac{\sigma^k}{k!} H_k(x,t)$, where

$$H_2 = x^2 - t, \quad H_3 = x^3 - 3tx, \quad H_4 = x^4 - 6tx^2 + 3t^2$$

are read off from the coefficients of $\sigma^2, \sigma^3, \sigma^4$ in $(1 + \sigma x + \frac{\sigma^2 x^2}{2} + \frac{\sigma^3 x^3}{6} + \frac{\sigma^4 x^4}{24})(1 - \frac{\sigma^2 t}{2} + \frac{\sigma^4 t^2}{8})$. Thus $\partial_\sigma^k M_t(\sigma)|_{\sigma=0} = H_k(B_t, t)$.

By (a), $\mathbb{E}[M_t(\sigma)\mathbf{1}_A] = \mathbb{E}[M_s(\sigma)\mathbf{1}_A]$ for $A \in \mathcal{F}_s$ and all σ. For $|\sigma| \le 1$, $\partial_\sigma^j M_t(\sigma) = P_j(B_t - \sigma t, t) M_t(\sigma)$ with P_j a polynomial. This is dominated by $C_j(1 + |B_t| + t)^j e^{|B_t|}$, which is integrable, so we may differentiate under $\mathbb{E}$ k times at $\sigma = 0$: $\mathbb{E}[H_k(B_t, t)\mathbf{1}_A] = \mathbb{E}[H_k(B_s, s)\mathbf{1}_A]$. Since $H_k(B_t, t)$ is adapted and integrable, it is a martingale. For $k = 2, 3, 4$ this gives the processes $B_t^2 - t$, $B_t^3 - 3tB_t$ and $B_t^4 - 6tB_t^2 + 3t^2$.